

# Aperture — Domain Model

**Version:** v1.0

**Document Type:** Domain Model

## Related Documents

- Product Requirements Document
  - System Architecture
  - Low-Level Design
  - Database Design
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## 1. Purpose

This document defines Aperture's business entities and the relationships between them.

Unlike the Database Design document, this document is **technology independent**.

It does not discuss:

- SQL
- indexes
- tables
- foreign keys
- caching

Instead, it answers:

What concepts exist in Aperture?

Which module owns them?

How do they relate?

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## 2. Domain Modeling Philosophy

Every entity should satisfy three rules.

## **Represents a business concept**

Examples:

- User
  - Movie
  - Review
- 

## **Has a clear owner**

Every entity belongs to exactly one module.

Other modules may reference it.

They should not own it.

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## **Has a single source of truth**

Avoid duplicate ownership.

Example

Movie

↓

Metadata Module

NOT

Search

NOT

Discovery

NOT

Streaming

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### 3. Core Domain Overview

The platform consists of nine primary domains.

``text id="bdt2g0" Identity



Users



Metadata



Reviews



Lists



Discovery



Streaming



Notifications



Administration ``

Every higher-level domain builds upon lower-level domains.

## 4. Identity Domain

Owned By

Authentication Module

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Primary Entity

Identity

Responsibilities

- authentication
- credentials
- providers
- sessions

Referenced By

- Users
- Administration

Never owns profile information.

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## 5. User Domain

Owned By

Users Module

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Primary Entity

User

Represents

A person's presence within Aperture.

Responsibilities

- profile

- preferences
- follows
- statistics

Referenced By

- Reviews
- Lists
- Discovery
- Notifications

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## 6. Metadata Domain

Owned By

Metadata Module

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Primary Entities

Movie

TV Show

Season

Episode

Person

Genre

Collection

Studio

Keyword

Represents

Every piece of cinematic content available on the platform.

Referenced By

Almost every other module.

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## 7. Review Domain

Owned By

Reviews Module

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Primary Entities

Review

Rating

Review Like

Represents

User opinions.

Reviews are considered first-class content.

Referenced By

Discovery

Notifications

Administration

Recommendation Engine

---

## 8. List Domain

Owned By

Lists Module

---

Primary Entities

List

List Item

Represents

Curated collections.

Lists are treated as independently discoverable content.

Referenced By

Discovery

Notifications

Recommendation Engine

---

## 9. Discovery Domain

Owned By

Discovery Module

---

Primary Entities

Discovery Feed

Recommendation Collection

Editorial Collection

Represents

Generated experiences.

Discovery owns orchestration.

Not persistent business data.

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## 10. Streaming Domain

Owned By

Streaming Module

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Primary Entities

Streaming Provider

Availability

Represents

Where content can currently be watched.

Streaming enriches metadata.

It does not replace it.

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## 11. Notification Domain

Owned By

Notifications Module

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Primary Entities

Notification

Preference

Represents

Communication with users.

Consumes events from other domains.

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## 12. Administration Domain

Owned By

Administration Module

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Primary Entities

Audit Record

Administrative Action

Feature Configuration

Represents

Operational capabilities.

Administration exists outside the normal business flow.

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## 13. Cross-Domain Relationships

Identity

↓

User

↓

Reviews

↓

Review Likes

---

User

↓

Lists



List Items



Movies

---

Movie



Reviews



Ratings



Recommendations

---

Movie



Streaming Availability

---

User



Notifications

---

These relationships define business semantics.

The Database Design document will define how they are persisted.

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## 14. Ownership Matrix

| Entity                 | Owner          | Referenced By                                |
|------------------------|----------------|----------------------------------------------|
| Identity               | Authentication | Users, Administration                        |
| User                   | Users          | Reviews, Lists, Discovery, Notifications     |
| Movie                  | Metadata       | Reviews, Lists, Search, Streaming, Discovery |
| TV Show                | Metadata       | Reviews, Lists, Search, Streaming            |
| Review                 | Reviews        | Discovery, Notifications                     |
| Rating                 | Reviews        | Discovery, AI                                |
| List                   | Lists          | Discovery, Notifications                     |
| Streaming Availability | Streaming      | Discovery                                    |
| Notification           | Notifications  | Users                                        |
| Audit Record           | Administration | Administration                               |

Ownership should remain exclusive.

References are encouraged.

Shared ownership is discouraged.

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## 15. Business Rules

Every entity should have:

- one owner
- one lifecycle
- one source of truth

Cross-domain updates should occur through service interfaces rather than direct modification.

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## 16. Domain Events

The following conceptual events exist within Aperture.

## Authentication

- UserRegistered
  - UserLoggedIn
- 

## Users

- ProfileUpdated
  - UserFollowed
- 

## Metadata

- MovieImported
  - MetadataUpdated
- 

## Reviews

- ReviewCreated
  - ReviewEdited
  - RatingChanged
- 

## Lists

- ListCreated
  - ListItemAdded
- 

## Streaming

- AvailabilityUpdated
- 

## Notifications

- NotificationCreated
- 

## Administration

- UserSuspended
- FeatureFlagUpdated

These events are conceptual within the modular monolith and provide a clear migration path toward asynchronous processing if required.

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## 17. Aggregate Boundaries

Each domain protects its own consistency.

Examples:

Movie

Owns:

- title
- runtime
- genres

Does not own:

- reviews
  - streaming
  - recommendations
- 

Review

Owns:

- content
- rating
- spoiler flag

Does not own:

- movie
- user

Only references them.

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List

Owns:

- ordering

- visibility

References:

- movies
  - TV shows
- 

## 18. Evolution Strategy

New business concepts should become new entities only when they represent distinct domain responsibilities.

Examples of future entities:

- Movie Space
- Badge
- Achievement
- Recommendation Explanation
- User Taste Profile
- Editorial Collection

Existing entities should evolve rather than duplicate responsibility.

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## 19. Interview Discussion Points

Be prepared to explain:

- Why separate Domain Model from Database Design?
  - Why does every entity have a single owner?
  - Why should Discovery not own Movies?
  - Why are Reviews first-class entities?
  - What are aggregate boundaries?
  - Why model conceptual domain events before introducing message queues?
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## 20. Summary

The Domain Model establishes the shared business vocabulary of Aperture.

It separates **what the platform represents** from **how it is implemented** and **how it is stored**, ensuring that business concepts remain stable even as the underlying architecture and persistence mechanisms evolve.

This document acts as the bridge between high-level architecture and detailed database design, providing a clear ownership model and consistent language for the rest of the engineering documentation.